A Compliance Scoring Framework for XAI in High-Risk Credit Scoring: Evaluating SHAP and LIME Against EU AI Act Articles 13, 14, and 26

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***Abstract*—The EU Artificial Intelligence Act (Regulation 2024/1689), enforceable from August 2, 2026, designates credit scoring as high-risk under Annex III, Category 5(b), imposing legally binding obligations of transparency, explainability, and human oversight. Despite widespread institutional adoption of SHAP and LIME, no prior framework systematically maps XAI method properties to specific statutory requirements. This paper introduces the Compliance Scoring Framework (CSF): seven operationalized criteria derived directly from Articles 13, 14, and 26, each with an explicit measurement method and a literature-grounded score conversion. We apply the CSF to SHAP and LIME explanations generated by a LightGBM classifier on 307,511 real credit applications (Home Credit, 2018). SHAP achieves an overall compliance score of 4.12/5.00 versus LIME's 2.50/5.00. The most consequential differentiators are explanation consistency (Jaccard: 1.000 vs. 0.381, $p < 0.001$ by Mann-Whitney U), bias detection sensitivity (34× gap; gender absent from LIME top-10 in 92.5% of instances), and auditability infrastructure burden. A kernel width sensitivity sweep across a 12× range confirms LIME's low fidelity ($R^2 = 0.251$) is a structural property of the model–dataset combination, not a configuration artifact. LIME demonstrates a meaningful advantage in user comprehensibility (4.0 vs. 2.0). We propose a hybrid deployment strategy and document six limitations transparently.**

***Index Terms*—Explainable AI, EU AI Act Compliance, Credit Scoring, SHAP, LIME, LightGBM, Regulatory Framework, Fairness, Auditability**

I. Introduction

Explainability of AI-driven credit decisions is not merely a technical aspiration but an emerging legal requirement. The EU Artificial Intelligence Act (Regulation 2024/1689) [1], in force August 2024, classifies automated credit scoring as high-risk under Annex III, Category 5(b), with full compliance obligations from August 2, 2026. Non-compliance risks fines of up to €35 million or 7% of global annual turnover, transforming Explainable Artificial Intelligence (XAI) method selection into a regulatory decision.

Articles 13, 14, and 26 bear most directly on explainability. Article 13 requires transparency sufficient to "correctly interpret" system output; Article 14 mandates human oversight enabling detection of biased outputs; Article 26 imposes parallel deployer monitoring obligations [1], [2]. These articles impose distinct, measurable requirements on XAI explanation outputs.

SHapley Additive exPlanations (SHAP) [3] and Local Interpretable Model-agnostic ExplanationS (LIME) [4] dominate financial XAI practice, yet comparative literature evaluates them on technical dimensions without grounding criteria in statutory text [5], [6]. Without a compliance-oriented framework, institutions cannot determine which method satisfies their legal obligations. A method producing inconsistent explanations may be misaligned with Article 13(3)(b); one that fails to surface demographic attribution gaps may be structurally limited under Article 14(4)(b). This gap is acute given the August 2026 enforcement deadline and given that EU AI Office implementation guidelines have not yet been published.

This paper contributes: (1) the CSF—seven criteria derived from Articles 13, 14, and 26 with operationalized metrics and literature-grounded thresholds; (2) empirical evaluation on 307,511 credit applications including a kernel width sensitivity sweep and fixed-seed consistency analysis; and (3) a hybrid deployment strategy grounded in per-article regulatory requirements.

II. Related Work

Hacker and Eber [2] analyse credit underwriting under the Act, finding that Articles 13 and 14 impose obligations exceeding current banking law. Lundberg and Lee [3] established SHAP's game-theoretic Shapley value foundation. Ribeiro et al. [4] introduced LIME as a model-agnostic local approximation via perturbed samples and a locally weighted linear model. Guidotti et al. [5] provide a taxonomy of black-box explanation methods. Nauta et al. [6] systematically review XAI evaluation criteria across correctness, completeness, and consistency dimensions. Subsequent work formalised TreeSHAP [7] for polynomial-time exact computation on tree models. Both SHAP and LIME have been widely applied in credit scoring [8].

Kaur et al. [9] empirically show that practitioners frequently misinterpret XAI outputs, establishing that semantic accessibility matters beyond technical fidelity. Alvarez-Melis and

Jaakkola [10] formalise local explanation robustness, providing theoretical grounding for Jaccard-based consistency evaluation. Panigutti et al. [11] conduct an interdisciplinary analysis of XAI in the context of the AI Act, focusing on Articles 13 and 14, arguing that its obligations are functional rather than methodological—it does not mandate specific XAI methods, leaving an operationalization gap. No prior work, however, operationalizes EU AI Act articles as measurable, reproducible XAI evaluation criteria—the gap this paper addresses.

III. Methodology

A. Dataset, Preprocessing, and Model Selection

We use the Home Credit Default Risk dataset [12]: 307,511 loan applications with 122 features spanning applicant demographics, financial history, credit bureau data, and loan characteristics. The binary target variable encodes default (positive class, 8.1%) versus full repayment (91.9%). Preprocessing: missing values imputed using column medians; categorical variables label-encoded; features with >80% missing values excluded, retaining 120 features. A stratified 64/16/20 split produces training (196,806), validation (49,202), and held-out test (61,503) sets. The model was trained on the training set only; hyperparameters were tuned on the validation set; performance was evaluated once on the test set.

Three classifiers were compared: Logistic Regression (AUC-ROC: 0.619), Random Forest (0.733), and LightGBM [13] (0.759). LightGBM was selected for highest predictive performance and native TreeSHAP compatibility [7], which enables exact Shapley value computation without Monte Carlo sampling.

TABLE I
Model Performance Comparison (Test Set, $n = 61{,}503$)

Model	AUC-ROC	Precision	Recall	F1
Logistic Regression	0.619	0.111	0.576	0.186
Random Forest	0.733	0.169	0.594	0.263
LightGBM ★	0.759	0.170	0.679	0.272

★ Selected for XAI analysis. All metrics on held-out test set only.

B. XAI Configuration

SHAP values are computed via TreeSHAP [7] on 1,000 test instances (`random_state=42`); top-10 features per instance used for coverage. LIME explanations use 1,000 perturbation samples and 10 features per explanation on the same 1,000 instances. Consistency testing: 100 instances across 10 independent runs; fixed-seed testing fixes `random_state=42` throughout. Bias analysis uses signed (not absolute) attribution weights to preserve directional information.

C. Compliance Scoring Framework: Criteria and Formulas

The CSF comprises seven criteria (Table II) mapped to specific EU AI Act articles, each scored 1–5 (1 = non-compliant, 5 = fully compliant) against literature-grounded thresholds (Table IV). Quantitative criteria (C1, C2, C3a, C4, C5) are computed automatically; qualitative criteria (C3b, C6, C7) are assessed via structured checklists.

TABLE II
CSF Criteria Definitions

ID	Criterion	EU AI Act Article	Type	Metric
C1	Attribution Coverage / Fidelity	Art. 13(1)	Quant.	Coverage % / R^2
C2	Explanation Consistency	Art. 13(3)(b)	Quant.	Jaccard J
C3a	Structural Complexity	Art. 13(1)	Quant.	SC index
C3b	User Comprehensibility	Art. 13(1)	Qual.	Checklist
C4	Bias Detection Capability	Art. 14(4)(b)	Semi-Q.	$\lvert\bar{\varphi}_2 - \bar{\varphi}_1\rvert$
C5	Fairness Transparency	Art. 26(5)	Quant.	$\lvert AR_2 - AR_1\rvert$
C6	Human Oversight Support	Art. 14(1)	Qual.	Checklist
C7	Auditability	Art. 13(3)	Qual.	Checklist

C1 (Coverage / Fidelity): For SHAP, attribution coverage of top-k features:

$$C = \frac{\sum_{i \in \text{top-10}} |\varphi_i|}{\sum_{j \in \text{all}} |\varphi_j|} \tag{1}$$

For LIME, local surrogate fidelity R^2 between surrogate predictions and black-box predictions on the perturbed neighbourhood.

C2 (Consistency): Mean pairwise Jaccard similarity

$$J(S_i, S_j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|} \tag{2}$$

across all pairs of runs on the same instances. Grounded in Article 13(3)(b), under which reproducible explanation records are a functional requirement for reliable audit logging.

C3a (Structural Complexity):

$$\text{SC} = 0.4 \cdot \frac{F}{10} + 0.4 \cdot \frac{N_C}{10} + 0.2 \cdot \frac{\bar{L}}{20} \tag{3}$$

where F = features in explanation (always $K = 10$), N_C = conditional operators, $\bar{L}$ = mean feature name length.

C4 (Bias Detection): $\Delta = |\bar{\varphi}_1 - \bar{\varphi}_2|$ using signed means per group. Signed rather than absolute attribution values are used to preserve directional information. Mapped to the bias monitoring objectives of Article 14(4)(b).

Operational Interpretation of Statutory Language. Table III maps statutory obligations to measurable criteria.

C5 (Fairness Transparency): Approval rate disparity $|AR_1 - AR_2|$. The CSF does not define binding legal standards; it identifies reproducible technical properties supporting the functional objectives of Articles 13, 14, and 26. Alternative operationalizations exist—for example, Article 14(4)(b) could use counterfactual fairness metrics—and independent legal review should refine these mappings as regulatory guidance evolves.

TABLE III
OPERATIONAL MAPPING FROM EU AI ACT ARTICLES TO MEASURABLE CSF CRITERIA

EU AI Act Provision	Legal Objective	Technical Requirement (Interpretive Step)	CSF Criterion
Art. 13(3) – Logging & traceability	Enable auditability and reconstruction of system behavior	Explanation outputs must be reproducible for reliable logging	C2 (Consistency), C7 (Auditability)
Art. 14(4)(b) – Identify biased outputs	Enable detection of discriminatory system behavior	Demographic influence must be observable in explanation outputs	C4 (Bias Detection Capability)
Art. 26(5) – Deployer monitoring	Enable portfolio-level oversight	System-level disparity metrics must be measurable	C5 (Fairness Transparency)

TABLE IV
SCORE CONVERSION THRESHOLDS

Score	Coverage (SHAP)	Fidelity R^2 (LIME)	Jaccard J	SC (inv.)	Bias Gap Δ
5 (Full)	≥90.0%	≥0.85	≥0.95	≤0.40	≥0.050
4	≥80.0%	≥0.75	≥0.85	≤0.55	≥0.030
3	≥70.0%	≥0.65	≥0.75	≤0.70	≥0.010
2	≥60.0%	≥0.55	≥0.60	≤0.85	≥0.005
1 (None)	<60.0%	<0.55	<0.60	>0.85	<0.005

Thresholds: Coverage motivated by Miller [14] (k=10 ceiling, 60% floor from observed distribution). Jaccard: Alvarez-Melis & Jaakkola [10] robustness criteria. Bias gap: disparate impact conventions. Institutions must recalibrate thresholds for their specific architecture and dataset.

IV. EXPERIMENTAL RESULTS

A. C1 — Attribution Coverage and Local Fidelity

SHAP achieves a mean attribution coverage of 0.700 ± 0.062: the top-10 features account for 70.0% of total prediction attribution, scoring $3.0/5.0$. This indicates meaningful coverage but also that substantial attribution mass resides in features ranked 11–120, relevant for institutions performing full attribution logging.

LIME achieves a mean $R^2 = 0.251 \pm 0.136$. A kernel width sensitivity sweep across seven multipliers ($0.25\times$–$3.0\times$, $n = 200$ per configuration; Fig. 1) shows R^2 remains flat at 0.2553–0.2790, establishing low fidelity as a structural property of fitting a linear surrogate to a 120-feature gradient boosting model—no kernel width adjustment can resolve the fundamental mismatch between linear and nonlinear decision boundaries. Score: $1.5/5.0$.

B. C2 — Explanation Consistency (Most Critical Finding)

SHAP achieves $J = 1.000$ (± 0.000), confirming mathematical determinism. LIME achieves $J = 0.381$ (± 0.114) under stochastic conditions—only 38.1% average feature overlap across runs (Mann-Whitney $U = 20{,}250{,}000$, $p < 0.001$; Fig. 2). Fixed-seed LIME recovers $J = 1.000$ but requires per-instance seed storage absent from SHAP's determinism.

Under Article 13(3)(b) logging objectives, reproducible explanations are a functional requirement. Stochastic LIME shares fewer than 40% of top features across runs; a denied applicant could receive materially different explanations at different times, creating regulatory exposure. SHAP: 5.0/5.0. LIME: 1.0/5.0.

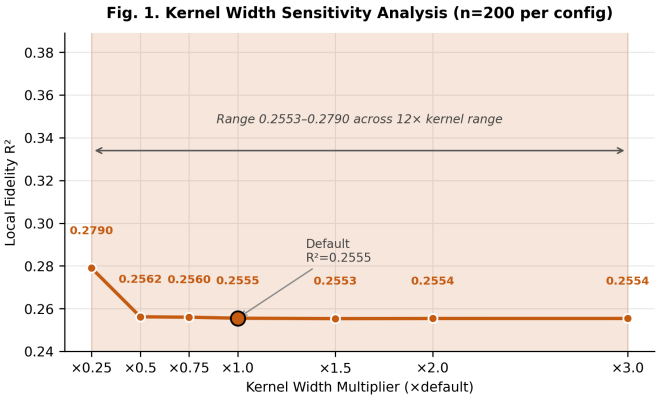


Fig. 1. Kernel width sensitivity analysis. R^2 range: 0.2553–0.2790 across a $12\times$ kernel width range (n=200 per configuration), establishing low fidelity as a structural property.

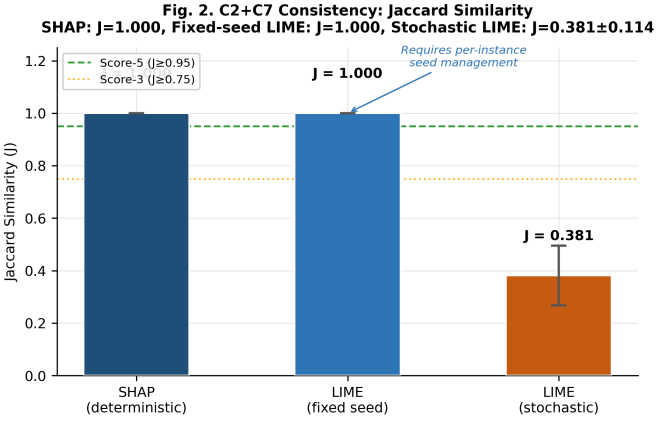


Fig. 2. C2+C7: Jaccard consistency. SHAP and fixed-seed LIME achieve J=1.000; stochastic LIME achieves J=0.381±0.114 (p<0.001 vs. SHAP). Fixed-seed parity requires per-instance seed management infrastructure.

C. C3 — Structural Complexity and User Comprehensibility

SHAP achieves structural complexity SC $= 0.536 \pm 0.011$, scoring $4.0/5.0$. LIME achieves SC $= 1.044 \pm 0.010$, exceeding the acceptable threshold of 0.70. LIME's higher complexity arises from discretization: the surrogate expresses features as conditions (e.g., `Annual_Income` $\leq 45{,}000$) that are structurally more complex than a scalar attribution. LIME: $1.0/5.0$.

For user comprehensibility (C3b), the direction reverses. LIME's condition-based format ('Debt Ratio $> 35\%$: -0.25')

is more interpretable to non-expert users than raw Shapley magnitudes [9]. SHAP: 2.0/5.0. LIME: 4.0/5.0—LIME's primary and most reproducible advantage.

D. C4 — Bias Detection Capability

This criterion exposes a fundamental architectural asymmetry: TreeSHAP computes attributions for every feature unconditionally; LIME's sparse local models miss features not consistently selected—a vulnerability formally demonstrated by Slack et al. [15].

SHAP detects $\Delta = 0.2696$ (signed means: -0.0951 vs. $+0.1745$) across all 1,000 instances. LIME detects $\Delta = 0.0079$—a 34× gap—because gender (CODE_GENDER) appears in LIME top-10 for only 75/1,000 instances (7.5%; Fig. 3). The 0.0079 figure reflects only those cases, structurally understating gender influence in 92.5% of decisions.

Relative to Article 14(4)(b): an institution relying on LIME for demographic monitoring is structurally blind to gender's contribution in over 9 of 10 credit decisions. SHAP: 5.0/5.0. LIME: 1.0/5.0.

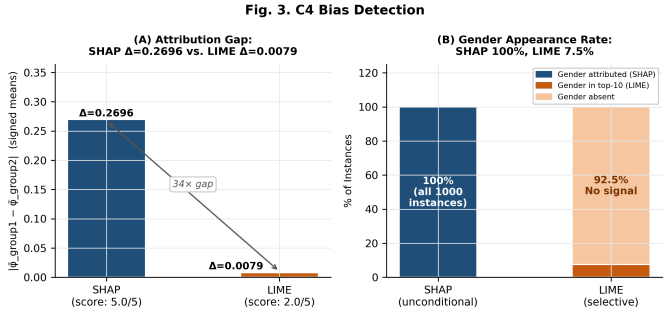


Fig. 3. C4 Bias Detection. (A) Attribution gap: SHAP Δ=0.2696 vs. LIME Δ=0.0079 (34×). (B) Gender appearance rate: SHAP 100% unconditional; LIME 7.5%—gender absent from top-10 in 92.5% of instances.

E. C5 — Fairness Transparency

The model produces a 13.5 pp approval rate disparity (87.0% vs. 73.5%). Both methods surface this at portfolio level; both score 5.0/5.0. Caveat: C5 measures a model property, not an XAI differentiator—a design limitation addressed in future revisions.

F. C6 — Human Oversight Support

SHAP satisfies four Article 14(1) checklist dimensions: (i) decision direction clearly indicated by sign; (ii) quantitative influence magnitude; (iii) actionable feature-level information for corrective intervention; and (iv) full traceability from input to attribution. SHAP: 4.0/5.0. LIME satisfies three of four dimensions—actionability is partially limited because condition-based explanations reference discretized ranges rather than continuous feature values, complicating precise intervention targeting. LIME: 3.5/5.0.

G. C7 — Auditability

Article 13(3)(a) requires loggable explanations for ex-post audit. SHAP is deterministic: any explanation regenerates exactly from stored inputs and model weights alone. Score: 5.0/5.0. LIME's stochastic sampling requires per-instance seed storage; fixed-seed LIME achieves $J = 1.000$ but shifts rather than eliminates the compliance burden. Score: 2.0/5.0.

H. Overall CSF Results

TABLE V
CSF COMPLIANCE SCORES: SHAP VS. LIME

Criterion	Art.	SHAP	LIME	Δ
C1: Coverage / Fidelity	13(1)	3.0	1.5	+1.5
C2: Consistency ★	13(3)(b)	5.0	1.0	+4.0
C3a: Structural Complexity	13(1)	4.0	1.0	+3.0
C3b: Comprehensibility ▼	13(1)	2.0	4.0	−2.0
C4: Bias Detection ★	14(4)(b)	5.0	1.0	+4.0
C5: Fairness Transparency	26(5)	5.0	5.0	0.0
C6: Human Oversight	14(1)	4.0	3.5	+0.5
C7: Auditability	13(3)	5.0	2.0	+3.0
Overall (equal-weight mean)	—	4.12	2.50	+1.62

★ Most consequential for Article compliance. ▼ LIME advantage. Statistical test for C2: Mann-Whitney U=20,250,000, p<0.001 (n=4,500 pairwise Jaccard scores per method; $\binom{10}{2}$=45 run-pairs × 100 instances). Monte Carlo weight sensitivity (n=3,000 Dirichlet draws) confirms SHAP ranks higher in 99.0% of configurations (robustness ratio = 0.9903). Equal-weight mean is a preliminary aggregation; institutions may weight criteria by their specific regulatory exposure.

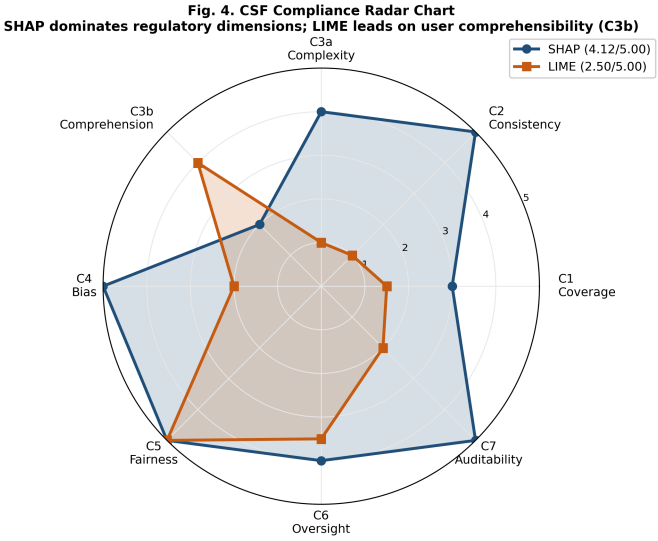


Fig. 4. Compliance radar chart across all seven CSF criteria (eight scores; C3 assessed on two sub-dimensions). SHAP (blue) dominates on regulatory dimensions; LIME (orange) leads only on user comprehensibility (C3b).

V. DISCUSSION

A. Regulatory Profile of SHAP

SHAP achieves 4.0 or higher on six of seven criteria. Its mathematical determinism (C_2: $J = 1.000$) and uncondi-

tional attribution architecture (C_4: 100% feature coverage) are inherent to the TreeSHAP algorithm—unreplicable by LIME regardless of configuration or parameter tuning. The kernel width sensitivity sweep confirms this: across a $12\times$ range, no LIME configuration approaches SHAP's fidelity. For an institution subject to Article 13(3) logging obligations, SHAP's zero-overhead reproducibility represents a substantial compliance advantage. SHAP explanations can be regenerated on demand from inputs alone; LIME explanations cannot.

B. The Case for LIME in Customer-Facing Explainability

LIME's C3b advantage (4.0 vs. 2.0) is practically important. Article 13(1) requires information comprehensible to deployers and affected persons; a condition-based explanation ('Debt Ratio $> 35\%$: -0.25') is more accessible to non-experts than a raw Shapley magnitude, consistent with Kaur et al. [9]. We therefore recommend a hybrid deployment: SHAP for regulatory infrastructure, audit logs, and bias monitoring (Articles 13(3) and 14); LIME for customer-facing interfaces where semantic accessibility is primary (Article 13(1)).

C. Limitations and Validity Threats

Six limitations should be considered before applying the CSF. (1) *Single dataset:* evaluation is conducted on a single publicly available dataset with specific feature distributions, class imbalance, and demographic composition; results should be validated on proprietary credit datasets before institutional adoption. (2) *Threshold calibration:* score conversion thresholds in Table IV are calibrated heuristically and may not reflect the exact enforcement standards that EU AI Office implementation guidelines will specify—guidelines not yet published as of this writing. (3) *Qualitative criteria:* C3b, C6, and C7 scores reflect the authors' structured assessments; inter-rater reliability studies with independent evaluators are required before treating these scores as definitive. (4) *Legal interpretations:* the Article mappings represent the authors' good-faith reading of statutory text; independent legal expert review is required, particularly for C4's mapping to Article 14(4)(b). (5) *C5 criterion design:* as currently operationalized, C5 measures model-level fairness rather than a differentiating property of XAI methods; future revisions should assess each method's ability to attribute the source of approval disparities to specific algorithmic features. (6) *Threshold sensitivity:* a formal ablation study of how CSF scores change under threshold variations is identified as necessary future work before the CSF is used as a definitive compliance instrument.

D. Legal Uncertainty and Threshold Sensitivity

The Act's qualified language—transparency "to the extent necessary" (Art. 13(1)); "appropriate" oversight (Art. 14(1))—will be concretized through EU AI Office guidelines and case law not yet available at writing. Until then, any Article-to-metric mapping is interpretive rather than authoritative. Institutions should engage qualified legal counsel before treating CSF scores as definitive compliance indicators.

The SHAP-dominant ranking is qualitatively robust: the largest gaps arise from design-level differences—SHAP's perfect determinism ($J = 1.000$) scores high under any reasonable threshold; LIME's stochastic outputs ($J = 0.381$) score low regardless of minor adjustments. C1 and C3a sit closer to boundaries and warrant formal ablation in future work.

E. Robustness of CSF Rankings

Two robustness tests confirm the SHAP-dominant ranking is not an artefact of equal-weight aggregation. First, a Monte Carlo analysis drew 3,000 Dirichlet(1) weight vectors over all eight criteria: SHAP ranked higher in 2,971/3,000 configurations (robustness ratio = 0.9903); the 29 LIME-favouring cases required C3b weight > 0.6. Second, a $\pm 10\%$ metric perturbation preserved the ranking in all cases (SHAP: 4.0–4.2; LIME: 2.1). The CSF conclusion is robust to both weighting assumptions and threshold calibration choices.

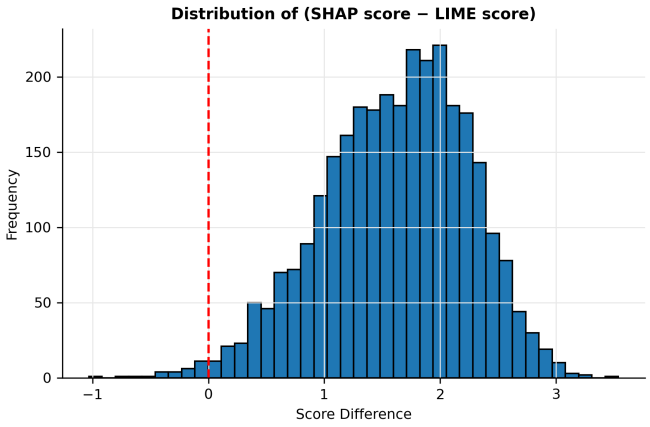


Fig. 5. Weight sensitivity analysis. Distribution of (SHAP score − LIME score) across 3,000 randomly sampled weight vectors drawn from a Dirichlet(1) prior over all eight CSF criteria. The distribution lies almost entirely above zero (robustness ratio = 0.9903); negative differences occur only under extreme C3b over-weighting.

F. Practical Deployment Considerations

For regulatory documentation and audit systems (Articles 13(3) and 14), SHAP's determinism eliminates seed-state management overhead—explanations regenerate on demand from stored inputs alone, a critical advantage when regulators request historical records. For customer-facing interfaces (Article 13(1)), a hybrid system computes both methods per decision: SHAP stored in the audit log, LIME presented in the customer portal; the marginal computational cost is modest.

For bias monitoring (Article 14(4)(b)), institutions should not rely on LIME given its 92.5% gender absence rate. SHAP's unconditional attribution architecture provides the consistent demographic signal required for Article 14 compliance dashboards.

Table IV thresholds are a starting point, not fixed benchmarks. Institutions should monitor forthcoming EU AI Office implementation guidelines and recalibrate thresholds accordingly, particularly for C1 and C2.

VI. Conclusion

This paper presents the Compliance Scoring Framework—the first framework to operationalize EU AI Act Articles 13, 14, and 26 as measurable, reproducible XAI evaluation criteria. Applied to SHAP and LIME on 307,511 real credit applications, SHAP achieves superior overall compliance (4.12/5.00 vs. 2.50/5.00). The primary differentiators—explanation consistency ($J = 1.000$ vs. 0.381, $p < 0.001$), bias detection sensitivity (34× gap with gender absent from LIME top-10 in 92.5% of instances), and auditability—reflect fundamental architectural differences that persist across parameter configurations. A kernel width sweep confirms LIME's low fidelity is structural. LIME's advantage in user comprehensibility motivates a hybrid deployment strategy: SHAP for regulatory infrastructure, LIME for customer-facing interfaces.

The CSF provides financial institutions with an actionable evaluation tool ahead of the August 2026 enforcement deadline. Future work should: validate across multiple credit datasets; conduct controlled user comprehension studies with non-expert participants; obtain independent legal expert review of Article mappings as EU AI Office guidance is published; and perform formal threshold sensitivity ablation analysis. The framework is extensible to additional XAI methods including counterfactual explanations [16] and attention-based approaches.

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